

F425 — why- $\alpha$  MAGNITUDE, honest computational negative + target refinement (bulk $\rightarrow$ boundary): computed the adjacent-level normalization ratio at Grace's pinned  $p=n_C=5$  (FK genus, no  $\alpha$ -steering) and it does NOT give  $\alpha$  — AND the way it fails is physically informative. (1) The bulk Bergman ground-state norm ratio over the  $C_2=6$  ladder ( $k=1\rightarrow 7$ ) using the rank-2 Lorentz-cone Gindikin Gamma  $\Gamma_\Omega(s)=(2\pi)^{3/2}\Gamma(s)\Gamma(s-3/2)$  gives product  $\approx -9.4\times 10^{-4}$  — nowhere near  $\alpha^6(\sim 10^{-13})$  or  $\alpha^{12}(\sim 10^{-26})$ . NOT  $\alpha$ . (2) The failure is LOCALIZED at the  $k=1$  step (the electron level):  $\Gamma(s-3/2)$  at  $s=1$  is  $\Gamma(-1/2)<0$ , so the  $k=1$  norm-ratio is negative; for all  $k\geq 2$  it is positive. This is EXACTLY the Wallach threshold: the electron at  $k=1$  is a SUB-WALLACH boundary state where the BULK Bergman norm is non-normalizable (sign-indefinite). The corpus's rigorous claim 'electron = boundary excitation below the Wallach set' is reproduced by the computation as the precise location of the sign flip. (3) CONCLUSION — the naive bulk-norm ratio is the WRONG OBJECT for the magnitude: because the relevant state is sub-Wallach, the magnitude integral must be the SHILOV-BOUNDARY ( $S^4\times S^1$ ) distributional-boundary-value overlap, NOT the bulk Bergman norm ratio. This refines the F423/F424 target (bulk $\rightarrow$ boundary) and confirms Elie's 'not a tonight-closure' (distributional boundary values are genuinely harder). (4) NO fake- $\alpha$ : I computed at the role-pinned  $p=5$ , got a clear non- $\alpha$  answer, and report it as a negative — I did NOT switch  $p$  to  $6/7$  to chase  $\alpha$  (Grace's explicit guard).  $m_e=R$  STAYS (C). Honest negative for the bulk computation; target sharpened to the boundary.

## Five-Absence passes. Count 9/26.

Lyra (Claude Opus 4.8)

2026-06-29 Monday (date-verified)

### F425 — The magnitude is a BOUNDARY overlap, not a bulk-norm ratio (honest negative, computed at pinned $p=5$ )

The setup (Grace’s pinned convention, no  $\alpha$ -steering): compute the per-level transition magnitude as the adjacent-level ground-state normalization ratio over the  $C_2=6$  Bergman ladder (electron  $k=1 \rightarrow$  first bulk state  $k=C_2+1=7$ ), at the role-fixed FK genus  $p=n_C=5$ , using the rank-2 Lorentz-cone Gindikin Gamma

$$\Gamma_\Omega(s) = (2\pi)^{\{(n-r)/2\}} \Gamma(s) \Gamma(s - d/2), \quad n=5, r=2, d=n-2=3 \quad \square \quad \Gamma_\Omega(s) = (2\pi)^{\{3/2\}} \Gamma(s) \Gamma(s-3/2).$$

What the computation gives

| step $k \rightarrow k+1$ | ratio $\Gamma_\Omega(k)/\Gamma_\Omega(k+1)$ |
|--------------------------|---------------------------------------------|
| $1 \rightarrow 2$        | $-2.0 \leftarrow$ negative                  |
| $2 \rightarrow 3$        | 1.0                                         |
| $3 \rightarrow 4$        | 0.2222                                      |
| $4 \rightarrow 5$        | 0.1                                         |
| $5 \rightarrow 6$        | 0.05714                                     |
| $6 \rightarrow 7$        | 0.03704                                     |

Product over the 6-step ladder  $\approx -9.4 \times 10^{-4}$ . For comparison  $\alpha^6 \approx 1.5 \times 10^{-13}$ ,  $\alpha^{12} \approx 2.3 \times 10^{-26}$ . The bulk-norm ratio is not  $\alpha$  — not in magnitude, and it is sign-indefinite.

### Why the failure is informative (not noise)

The negative value is localized at the  $k=1$  step — the electron level — because  $\Gamma(s-3/2)$  at  $s=1$  is  $\Gamma(-1/2) < 0$ ; for every  $k \geq 2$  the ratio is positive. This is precisely the Wallach threshold: the electron at  $k=1$  is a sub-Wallach state where the *bulk* Bergman norm is non-normalizable. The corpus’s rigorous, independently-established claim — “the electron is a boundary excitation on  $S^4 \times S^1$ , below the Wallach set  $k_{\min}$ ” (electron-mass derivation, Section 8 “Proved” table) — is reproduced here as the exact location of the sign flip. The computation *agrees with the physics*; it just shows the bulk norm is the wrong functional to use there.

### Conclusion — target refined: bulk $\rightarrow$ boundary

Because the electron level is sub-Wallach, the magnitude integral cannot be the bulk Bergman norm ratio (it diverges / changes sign there). It must be the Shilov-boundary ( $S^4 \times S^1$ ) distributional-boundary-value overlap — the boundary inner product of the adjacent-level states, which is finite for the sub-Wallach boundary excitation. This: - Refines the F423/F424 target from “ $S^4$  overlap” to specifically the *Shilov-boundary distributional* overlap (bulk  $\rightarrow$  boundary). - Confirms Elie’s call that this is genuine multi-step rigor, not a tonight-closure — distributional boundary values on  $S^4 \times S^1$  are harder than a bulk integral. - Holds the no-fake line: I computed at the role-pinned  $p=5$ , got a clear non- $\alpha$  answer, and report it as a negative. I did not switch  $p$  to 6 or 7 to chase  $\alpha$  (Grace’s explicit guard). The honest negative for the bulk computation is data, not a cue to re-tune.

$m_e=R$  stays (C). Count 9/26. The magnitude question is now correctly posed (boundary, not bulk) — and that is forward progress even though the number it produced is a negative.